

Infinitely Many Integer Solutions of $x^3 + x^2y^2 + z^2 + 1 = 0$

1 A short proof of infinitude

Put

$$Q(t) = 465577896 + 1881226506t + 1900333542t^2.$$

The leading coefficient is positive and nonsquare, the discriminant is $-1853382492 \neq 0$, and

$$Q(-659) = 28706088^2.$$

Hence Gauss's theorem on generalized Pell equations implies that $Q(t)$ is a square for infinitely many integers t . For every such t , choose either integer square root y of $Q(t)$, and set

$$\begin{aligned} x &= -577630421 - 2333988934t - 2357694538t^2, \\ z &= 50421655389668t^3 + 74872031013786t^2 \\ &\quad + 37059612550518t + 6114499038718. \end{aligned}$$

Direct expansion gives the polynomial identity

$$x(t)^3 + x(t)^2Q(t) + z(t)^2 + 1 \equiv 0,$$

so every such t gives an integer solution. Since the admissible integers t are infinite and the displayed polynomial $x(t)$ is a nonconstant quadratic, infinitely many distinct solutions result.

Here Gauss's theorem is used in the standard form: if $A > 0$ is nonsquare, $B^2 - 4AC \neq 0$, and $Y^2 = AT^2 + BT + C$ has one integral solution, then it has infinitely many integral solutions.

2 How the parametrization was found

This section explains the discovery process only. No additional proof is needed, because the theorem above is already certified by Gauss's theorem and the displayed polynomial identity.

1. The Gaussian-norm clue

Writing $x = -a$ changes the equation into

$$z^2 + 1 = a^2(a - y^2).$$

The left-hand side is the Gaussian norm of $z + i$. A small exact search first found

$$(x, y, z) = (-493, 12, 9210).$$

This solution has

$$493 = 22^2 + 3^2, \quad 493 - 12^2 = 349 = 18^2 + (-5)^2,$$

and, more revealingly,

$$(22 + 3i)^2(18 - 5i) = 9210 + i.$$

That identity suggested forcing, in general,

$$(p + iq)^2(r + is) = z + i.$$

Its real and imaginary parts lead to the search conditions

$$a = p^2 + q^2, \quad a - y^2 = r^2 + s^2, \quad 2pqr + (p^2 - q^2)s = 1,$$

with

$$z = (p^2 - q^2)r - 2pqs.$$

Thus the problem became one of finding a one-parameter integral line on

$$2pqr + (p^2 - q^2)s = 1$$

for which $p^2 + q^2 - r^2 - s^2$ is a quadratic having one square value.

2. Solving for an integral line

The search used linear polynomials

$$p(t) = At + B, \quad q(t) = Ct + D, \quad r(t) = Et + F, \quad s(t) = Gt + H.$$

Once A, B, C, D are fixed and $\delta = AD - BC \neq 0$, equating the four coefficients in

$$2p(t)q(t)r(t) + (p(t)^2 - q(t)^2)s(t) \equiv 1$$

gives the following exact recipe:

$$\begin{aligned} E &= \frac{(A^2 - C^2)(A^2 + C^2)}{2\delta^3}, & G &= -\frac{AC(A^2 + C^2)}{\delta^3}, \\ F &= \frac{AB(A^2 + 3C^2) - CD(3A^2 + C^2)}{2\delta^3}, & H &= -\frac{C^3B + A^3D}{\delta^3}. \end{aligned}$$

This converts the apparently unstructured search for eight coefficients into a much smaller divisibility search for four coefficients.

A convenient scaling is

$$A = L^2u, \quad C = L^2v, \quad \delta = L^3.$$

Then the first two coefficients simplify to

$$E = \frac{(u^2 - v^2)(u^2 + v^2)}{2L}, \quad G = -\frac{uv(u^2 + v^2)}{L}.$$

The search therefore enumerated small u, v, L , preferred values for which $L \mid u^2 + v^2$, solved the determinant equation, imposed the remaining integrality conditions, and finally tested the resulting quadratic for one square value.

3. The origin of the large coefficients

One successful arithmetic choice was

$$u = 23, \quad v = -27, \quad L = 37,$$

noting that

$$23^2 + 27^2 = 1258 = 34 \cdot 37.$$

It gives

$$A = 37^2 \cdot 23 = 31487, \quad C = -37^2 \cdot 27 = -36963.$$

The determinant requirement $AD - BC = 37^3$ becomes the small linear equation

$$23D + 27B = 37.$$

Its complete integral solution is

$$B = 15 + 23k, \quad D = -16 - 27k.$$

Substitution into the coefficient recipe above gives

$$E = -3400, \quad G = 21114,$$

and

$$F = -\frac{2(1700k + 429)}{1369}, \quad H = \frac{21114k + 13241}{1369}.$$

The requirement that both F and H be integers reduces to

$$k \equiv 677 \pmod{1369}.$$

Taking the smallest nonnegative representative, $k = 677$, produces

$$B = 15586, \quad D = -18295, \quad F = -1682, \quad H = 10451.$$

Thus the four linear polynomials are

$$\begin{aligned} p(t) &= 31487t + 15586, \\ q(t) &= -36963t - 18295, \\ r(t) &= -3400t - 1682, \\ s(t) &= 21114t + 10451. \end{aligned}$$

In particular, the numbers 31487 and 15586 were not guessed:

$$31487 = 37^2 \cdot 23, \quad 15586 = 15 + 23 \cdot 677.$$

4. The final square-value filter

For these four lines, exact expansion gives

$$p(t)^2 + q(t)^2 - r(t)^2 - s(t)^2 = 465577896 + 1881226506t + 1900333542t^2 = Q(t).$$

An exact integer scan found the square value

$$Q(-659) = 28706088^2.$$

At that point Gauss's theorem supplies infinitely many further square values automatically. Finally, expanding

$$x(t) = -(p(t)^2 + q(t)^2), \quad z(t) = (p(t)^2 - q(t)^2)r(t) - 2p(t)q(t)s(t)$$

produces exactly the two polynomials used in the short proof.

The discovery was therefore a combination of a conceptual reduction (Gaussian norms), exact symbolic coefficient solving, and a bounded integer search for divisibility and one initial square value. The large coefficients are the expanded output of that structured procedure, not unmotivated constants.